

Customer, Product, and Profitability Performance Analysis in Supply Chain Operations

Executive Summary | APL Logistics (KWE Group) | Prepared by Isha Kalra, 2026

BUSINESS PROBLEM

APL Logistics generates extensive order, customer, product, and pricing data, but leadership has lacked visibility into which customers, products, and markets actually generate profit rather than just revenue. High sales volume, active discounting, and broad market reach can mask underlying margin erosion and loss-making relationships. This project delivers financial clarity and margin intelligence to support profit-focused, rather than purely revenue-focused, decision-making.

SCOPE OF ANALYSIS

The analysis covers **180,519 order-item records across 40 variables**, spanning 20,652 unique customers, 118 products, 50 product categories, 5 markets, and 23 order regions. The dataset was validated for missing values, duplicates, and financial consistency before profitability analysis, ensuring the findings below rest on a reliable data foundation.

Total Sales	Total Profit	Profit Margin	Loss-Making Customers
36.78M	3.97M	10.78%	19.70%

KEY FINDINGS

- Revenue does not equal profit. The business generated 36.78 million in sales and 3.97 million in profit — an overall margin of 10.78% — but this aggregate figure hides sharp differences across customers, products, and markets.
- Nearly 1 in 5 customers loses money. 4,069 customers (19.70% of the base) are individually loss-making, despite generating 19.34% of total sales — revenue that does not translate into value.
- Profit is concentrated in very few products. The top 10 products generate 89.70% of total profit, while the top 10 customers contribute only 0.50% — product and customer strategy require fundamentally different approaches.
- Discounting steadily erodes margin. Profit margin falls from 13.09% on non-discounted orders to 9.39% at 20–25% discount, a 3.71 percentage-point decline, though loss rates do not rise in step with discount depth.
- Market scale and market efficiency are different things. Europe leads in total sales and profit, but USCA delivers the highest margin (11.14%), while Pacific Asia trails on both margin (10.37%) and loss rate (18.92%).

RECOMMENDATIONS

- Introduce margin-aware discount governance, with added scrutiny above the 20–25% discount band and category-specific controls for weak-margin categories such as Accessories and Men's Footwear.
- Review the 4,069 loss-making customers for pricing, discount exposure, and service cost before further investment, while continuing to protect high-value customer relationships.
- Closely monitor the small set of products generating most profit, and investigate the six identified high-revenue, low-margin products for pricing or cost-structure review.
- Apply differentiated market strategy: protect Europe's scale, replicate USCA's margin efficiency where possible, and prioritize Pacific Asia for profitability review.

DASHBOARD AND DECISION-SUPPORT VALUE

An interactive Streamlit dashboard operationalizes these findings — Revenue & Profit Overview, Customer Value, Product & Category Performance, Discount Impact Analyzer, and Market & Regional Analysis modules, each filterable by segment, category, product, market, region, and discount rate. This provides an interactive, decision-ready view of profitability intelligence, enabling APL Logistics to manage not just how fast it delivers, but how profitably it operates.